AGILE LAB FAT

QUESTION:SMART ELECTRICITY BILLING SYSTEM

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Smart Electricity Billing System
Problem Statement:

Ensure electricity usage is calculated and billed correctly.
Tasks:

1)Develop billing calculation logic

2)Write JUnit tests for tariff calculations

3)Implement CI/CD pipeline with automated testing

• Set up version control using GitHub

Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commitsAutomate build process using MavenIntegrate automated unit tests in pipeline

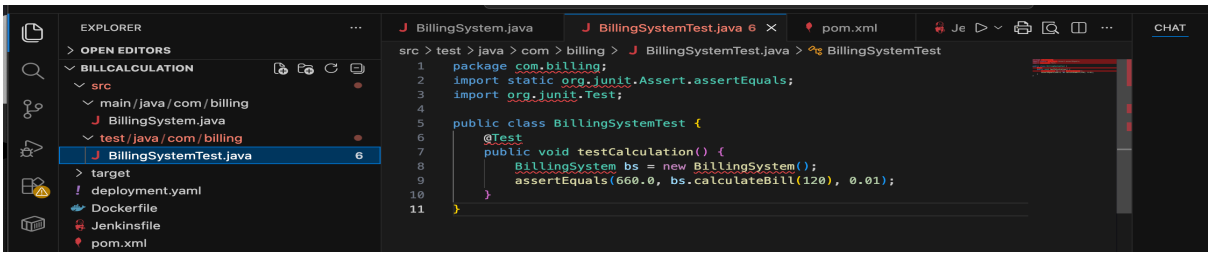
◦ Configure CD pipeline for automatic deployment to staging/production

4) Deploy services on Docker and Kubernetes

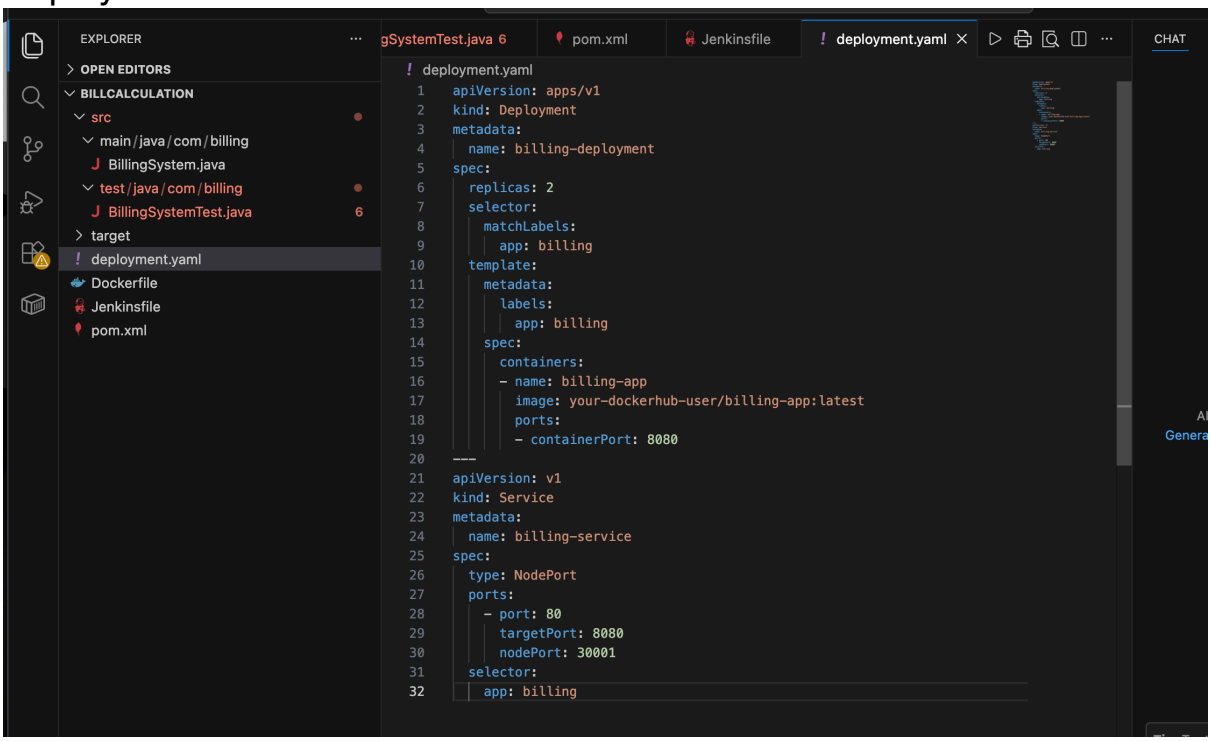
1) Develop billing calculation logic

Code:
Java code:

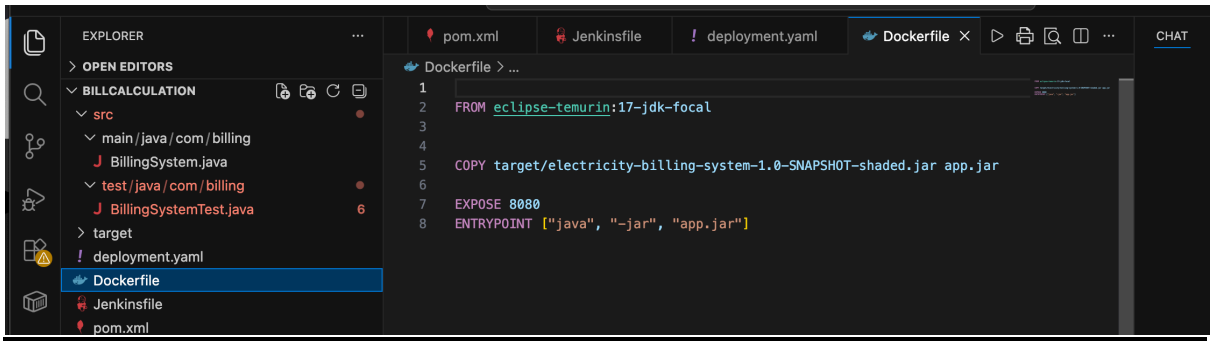
2)Junit test:



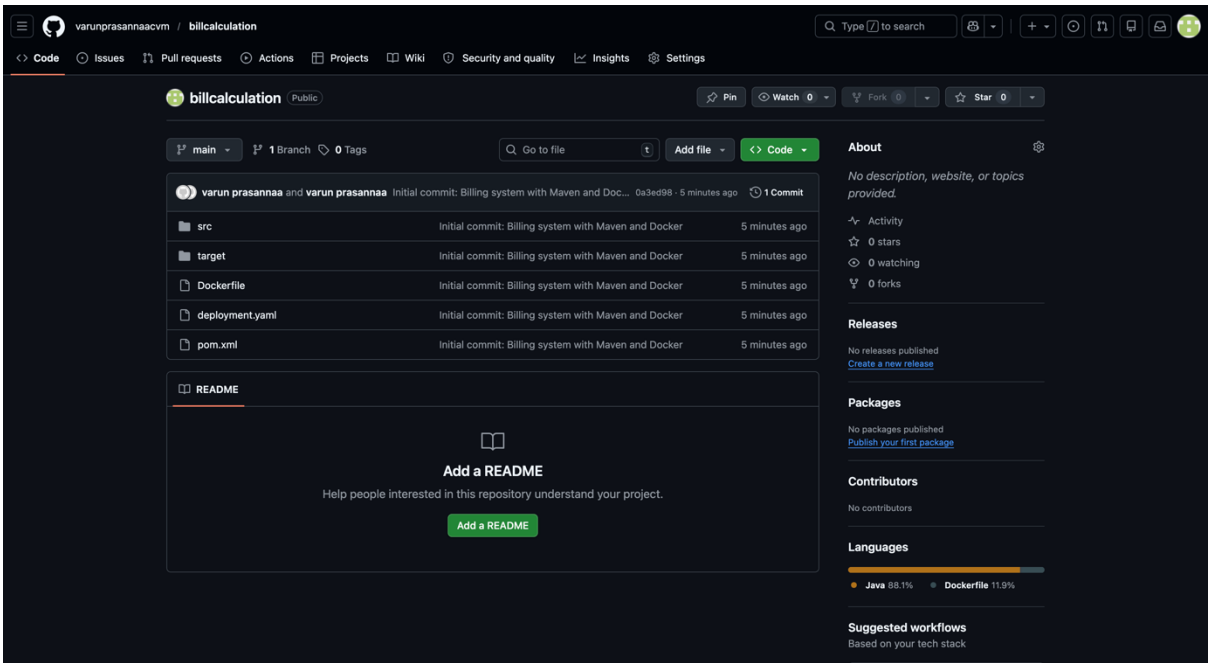
Deployment:

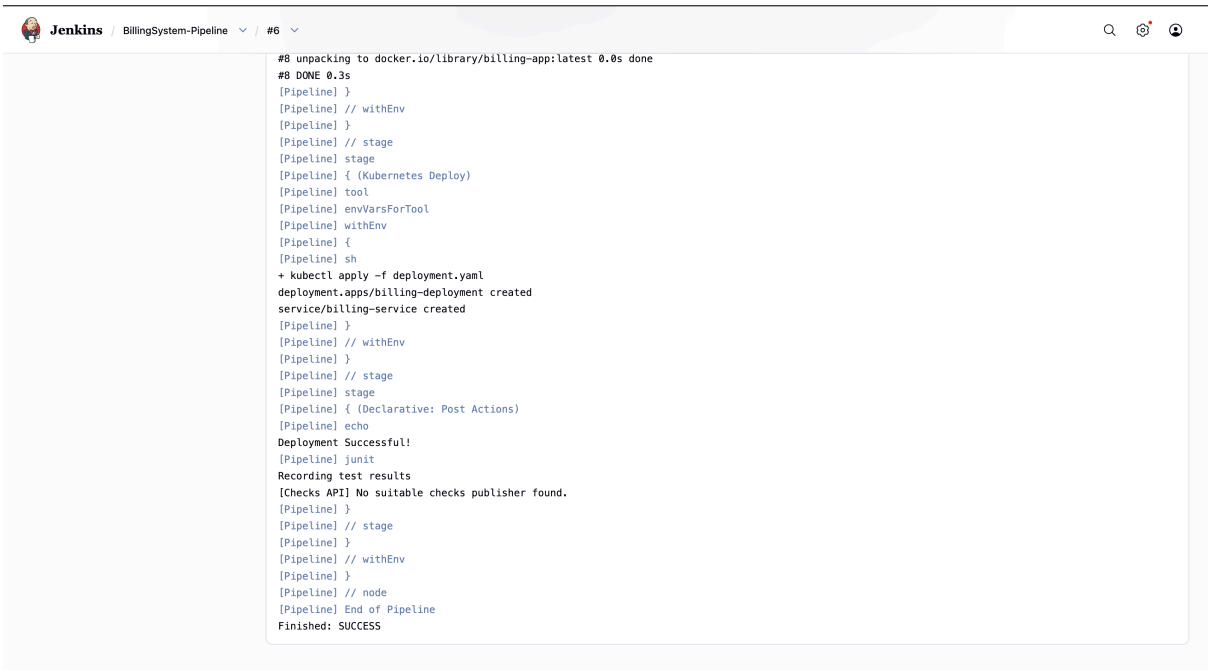


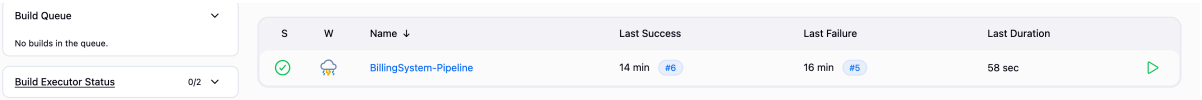
Docker:



3) Implement CI/CD pipeline with automated testing:

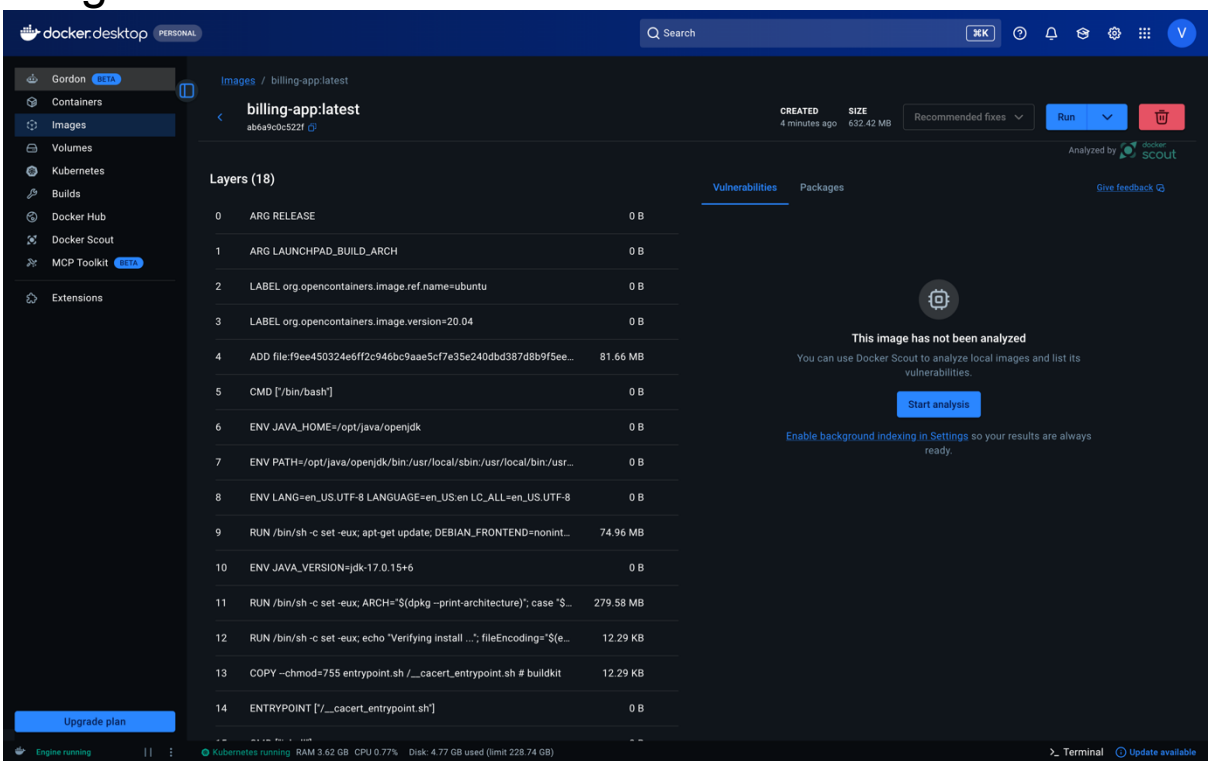


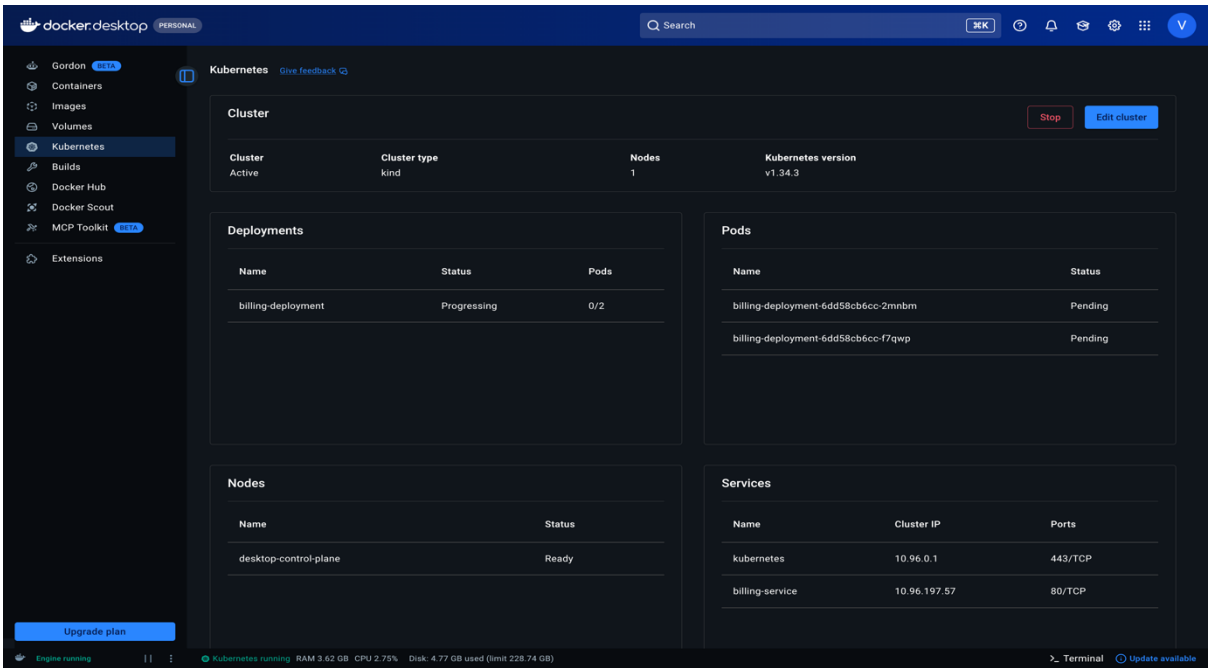




4. Deploy services on Docker and Kubernetes:

Images:



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Last login: Wed Apr 15 16:31:22 on ttys000
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get pods
No resources found in default namespace.
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get service billing-service
Error from server (NotFound): services "billing-service" not found
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get pods
NAME                                  READY   STATUS         RESTARTS   AGE
billing-deployment-6dd58cb6cc-2mnbm   0/1     ErrImagePull   0          2m2s
billing-deployment-6dd58cb6cc-f7qwp   0/1     ErrImagePull   0          2m2s
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get service billing-service
NAME              TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
billing-service   NodePort   10.96.197.57   <none>        80:30001/TCP   2m9s
varunprasannaa@varuns-MacBook-Pro ~ %
```